

INTERMEDIATE-CARDINALITY STRATA AND DIRECT LIMITS IN LEAN: A FORMALIZATION STUDY AROUND THE CONTINUUM

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ABSTRACT. This paper draft presents a Lean 4 formalization study of an Intermediate-Cardinality Arithmetic Hypothesis (ICAH): a stratified view of the real continuum under the negation of the Continuum Hypothesis. The development packages intermediate-size subsets of $\mathbb{R}$ as strata, bundles cardinal data with ordered-field structure through size-aware fields, and studies elementary chains and their direct limits in the language of ordered rings. The formalization proves, among other results: that $\mathbb{R}$ is a real-closed field; that for every cardinal $\aleph_0 \leq \kappa \leq 2^{\aleph_0}$ there is a real-closed subfield of $\mathbb{R}$ of cardinality exactly κ ; a Łoś-style elementarity theorem for direct limits of elementary chains compatibly embedded in $\mathbb{R}$; a cardinal-arithmetic theorem for such direct limits; and, under $\neg \text{CH}$, a continuum-indexed cofinal family of intermediate-size real-closed subfields whose union is $\mathbb{R}$. The assembled main theorem depends on exactly two named axioms — the set-theoretic regime $\neg \text{CH}$ and the model completeness of real-closed fields (Tarski–Seidenberg), the latter being the single remaining Mathlib gap — and the axiom inventory is machine-checked in the build.

WORKING DRAFT · version 0.3.0 · June 2026

1. INTRODUCTION

The Continuum Hypothesis (CH) asserts that there is no cardinal strictly between the cardinality of the natural numbers and the cardinality of the continuum. Consequently, any mathematical program that treats intermediate cardinalities inside $\mathbb{R}$ must either work relative to $\neg \text{CH}$, change the meaning of size, or explicitly track which assumptions are external to ZFC.

The Lean development studied here chooses the first route. It introduces a named set-theoretic assumption, `ICAH.not_CH : continuum  $\neq$  aleph 1`, and uses it to witness an intermediate cardinal. Around this assumption, the project builds a formal architecture for strata of the real continuum, size-aware field structures, subfield refinements, definability kernels, elementary chains, and direct limits.

Contribution. The main contribution of the current draft is not a new set-theoretic independence theorem. It is a formalization architecture: a Lean-readable decomposition of a continuum-stratification hypothesis into proved lemmas, named assumptions, and Mathlib-facing gaps. After the latest axiom-reduction stage, the assembled main theorem `icahTheorem` depends on exactly two named axioms: the set-theoretic regime $\neg \text{CH}$ and the model completeness of real-closed fields (Tarski–Seidenberg), the latter being the single remaining Mathlib gap.

The project should be positioned for a mathematical audience as a formalization paper with three layers:

1. a set-theoretic layer, where $\neg \text{CH}$ produces intermediate cardinal witnesses;
2. an algebra/model-theoretic layer, where real-closed subfields of $\mathbb{R}$ of every infinite cardinality up to the continuum are constructed and assembled into elementary chains;
3. a proof-engineering layer, where the one remaining dependency is exposed as a named axiom, machine-audited in the build, and can be attacked as an independent Mathlib contribution.

The present paper draft is organized as follows. Section 2 recalls the mathematical background. Section 3 describes the Lean architecture. Section 4 isolates the results that are already proved. Section 5 gives the axiom inventory and explains which parts are mathematical assumptions and which parts are formalization gaps. Section 6 discusses related work. Section 7 sketches analogues in valued and p-adic field theory. Section 8 concludes.

1.1. Guiding problem.

Definition. An *intermediate-cardinality stratum* is, informally, a subset $S \subseteq \mathbb{R}$ equipped with a cardinal κ such that

$$\aleph_0 < \kappa < 2^{\aleph_0}$$

and a proof that the subtype determined by S has cardinality κ .

Definition. A *size-aware field* is a bundled field-like object carrying its underlying type, a designated cardinal, and the typeclass instances needed to treat the carrier as an ordered ring or field in Lean.

The central formal question is:

Question. Can one build, under explicit assumptions, a hierarchy of intermediate-size strata whose carriers support enough algebra and model theory to form elementary chains with a direct limit of cardinality continuum?

The current Lean code answers this question constructively up to a single model-theoretic assumption: every clause of the formal statement is proved once the model completeness of real-closed fields is granted.

2. BACKGROUND

2.1. The continuum and the role of $\neg \text{CH}$.

Let $c = 2^{\aleph_0}$ denote the cardinality of the continuum. The Continuum Hypothesis states that there is no cardinal κ satisfying $\aleph_0 < \kappa < c$. Since Gödel and Cohen, the standard mathematical background is that CH is independent of ZFC, assuming ZFC is consistent [1, 2].

The Lean project therefore does not attempt to prove $\neg \text{CH}$ in ZFC. Instead, it introduces $\neg \text{CH}$ as a named axiom and develops all downstream objects relative to that assumption.

Proposition. Under the project axiom `ICAH.not_CH`, the cardinal $\aleph_1$ witnesses the existence of an intermediate cardinal:

$$\exists \kappa, \aleph_0 < \kappa \wedge \kappa < 2^{\aleph_0}.$$

The Lean proof uses the standard inequality $\aleph_0 < \aleph_1$, the theorem $\aleph_1 \leq c$, and converts the inequality plus non-equality $c \neq \aleph_1$ into the strict inequality $\aleph_1 < c$.

2.2. Real closed fields and definability.

Real closed fields are the algebraic and model-theoretic setting closest to the ordered real field. They admit quantifier-elimination results and provide the natural language in which order, addition, multiplication, and definability can be studied together [3, 4].

ICAH uses this background in a restrained way. The code does not merely assume every bare subset of $\mathbb{R}$ is a field. Instead, it introduces `SubfieldStratum`, which requires the carrier set to be the underlying set of a `Subfield RR`. This is the right formal move: closure under field operations is obtained from the subfield structure rather than reconstructed ad hoc.

2.3. Elementary chains and direct limits.

An elementary chain is a sequence of structures connected by elementary embeddings. The classical model-theoretic expectation is that unions or directed limits of elementary chains preserve the relevant first-order theory under suitable hypotheses [5].

The Lean project packages this through `ElemChain`, where each level carries an `LOR`-structure and each successor map is an elementary embedding. The direct limit is then implemented using Mathlib's `Language.DirectLimit` API.

3. FORMAL ARCHITECTURE

This section records the paper-level interpretation of the Lean objects. The goal is to make the repository legible to mathematicians without forcing them to read Lean code first.

3.1. Core objects.

Definition. `Stratum` is the basic continuum-layer object. It consists of an ordinal index, a set $S \subseteq \mathbb{R}$, a cardinal κ , a witness that the subtype determined by S has cardinality κ , and bounds $\aleph_0 < \kappa < c$.

Definition. `SizeAwareField` is a bundled structure containing a carrier type, a designated cardinal, and the instances `[Field K]`, `[LinearOrder K]`, and `[IsStrictOrderedRing K]`. This avoids relying on deprecated or overly strong ordered-field classes while still exposing enough structure for model-theoretic use.

Definition. `SubfieldStratum` refines `Stratum` by adding a subfield `subfield : Subfield RR` and a proof that the stratum set is exactly the underlying set of that subfield.

The role of `SubfieldStratum` is central. A raw set of real numbers is not automatically closed under addition, multiplication, negation, and inverses. A subfield is. Thus, the field part of the theory is moved from a fragile closure proof obligation into a stable algebraic structure.

Definition. `RCSubfieldStratum` refines `SubfieldStratum` once more by requiring the subfield to be **real closed**. This is the correct hypothesis for model theory: a general subfield of $\mathbb{R}$ (such as $\mathbb{Q}$) is not elementarily embedded in $\mathbb{R}$, while a real-closed subfield is, by model completeness of the theory of real-closed fields.

Construction. `subfieldToSAF` converts a subfield of $\mathbb{R}$, together with a cardinality witness, into a `SizeAwareField`. The key design choice is to use subtype inheritance for the linear order and Mathlib’s subfield ordered-ring instance for the strict ordered ring structure.

Construction. `relAlgebraic K0` is the relative algebraic closure of a subfield $K_0 \subseteq \mathbb{R}$ inside $\mathbb{R}$. It is proved real closed via a root-closure criterion, and its cardinality is bounded by the maximum of the cardinality of K_0 and $\aleph_0$. This is the engine behind `exists_rc_subfield`, which produces a real-closed subfield of any prescribed infinite cardinality up to the continuum.

3.2. Definability layer.

The formalization uses an abbreviation `LOR` for the first-order language used in the definability kernel. In the current development, the graphs of addition and multiplication on $\mathbb{R}$ are proved definable by constructing bounded formulas and transporting realization through Mathlib’s language-homomorphism machinery.

Remark. The definability results are important because they demonstrate that the paper is not only doing cardinal bookkeeping. It also touches the first-order model-theoretic infrastructure needed to express arithmetic inside a layer.

3.3. Elementary chains.

Definition. `ElemChain` packages a sequence `obj : ℕ → Type*` of first-order structures and elementary embeddings `obj n ↪e obj (n+1)`.

From these successor maps, the project defines two related systems:

1. `embLE`, which composes successor elementary embeddings and preserves elementarity;
2. `sysEmb`, which uses Mathlib’s directed-system API to build the underlying embedding system.

The lemma `embLE_eq_sysEmb` proves that these two constructions agree as functions. This is a small but important bridge: the proof-relevant elementary embedding API and the directed-colimit API are not automatically the same object.

3.4. Direct limit.

Definition. For an elementary chain $\mathbb{C}$, the direct limit `DirectLim C` is defined as `Language.DirectLimit C.obj (...)`, using the underlying directed system of embeddings.

The direct limit is the formal version of the intended limit field F_ω . The project proves both a substantial cardinal theorem for this direct limit (`directLimit_card`) and its elementarity over $\mathbb{R}$ (`losDirectLimit`, via the Tarski–Vaught test); both were previously exposed as gaps and are now theorems.

3.5. Cofinal family at the continuum.

The $\mathbb{N}$ -indexed chain cannot, by itself, exhaust $\mathbb{R}$: König’s theorem gives $\text{cof}(c) > \aleph_0$, so a countable increasing union of sets of size $< c$ has size $< c$. The module `ICAH/CofinalFamily.lean` therefore introduces a `ℵ.ord`-indexed monotone family of intermediate-size real-closed subfields whose union is all of $\mathbb{R}$ (`exists_cofinal_rc_family`). The top-level statement `ICAHStatement` uses this family for its limit-size clause, which keeps the formalized claim non-vacuous.

4. PROVED RESULTS

This section separates fully proved Lean results from assumptions or future work. In the final paper, each result should include a pointer to the Lean declaration and, when appropriate, a compressed proof sketch.

4.1. Intermediate cardinal witness.

Theorem. `exists_intermediate_cardinal`: assuming `ICAH.not_CH`, there exists a cardinal κ such that $\aleph_0 < \kappa < c$.

Remark. The witness is $\aleph_1$. The proof uses `aleph0_lt_aleph_one`, `aleph_one_le_continuum`, and turns non-equality into strict inequality through `lt_of_le_of_ne`.

4.2. Concrete synthetic stratum.

Construction. `syntheticStratum` materializes an intermediate-cardinality subset of $\mathbb{R}$ by extracting a concrete set $S \subseteq \mathbb{R}$ with cardinality $\aleph_1$ from a cardinal existence theorem.

This is the first bridge from a cardinal statement to an inhabited Lean structure. It turns the abstract existence of an intermediate cardinal into a concrete `Stratum` object.

4.3. Subfield strata and size-aware fields.

Theorem. `fieldOnSubfieldStratum`: every `SubfieldStratum` yields a `SizeAwareField` whose carrier and cardinal match the underlying stratum.

The technical point is that the carrier equality is treated as a propositional equality of subtypes, not merely a bijection. This is precisely what the `SizeAwareField.carrier` field requires.

Theorem. `fieldOnStratum`: every `Stratum` (not only subfield strata) admits a `SizeAwareField` with matching carrier and cardinal. Formerly an axiom.

The proof transports the field structure of a real-closed subfield of the right cardinality (from `exists_rc_subfield`, below) across a cardinality `Equiv`, using `Equiv.field` and `LinearOrder.lift`.

4.4. Real-closedness of $\mathbb{R}$ and of relative algebraic closures.

Theorem. `Real.isRealClosed`: $\mathbb{R}$ is a real-closed field. Formerly an axiom.

Remark. The proof uses `IsRealClosed.of_linearOrderedField`: squares are handled by `Real.sqrt`, and odd-degree polynomials acquire roots by the intermediate value theorem combined with the leading-coefficient asymptotics `Polynomial.tendsto_atTop/atBot`.

Theorem. `isRealClosed_of_forall_root` and `relAlgebraic_isRealClosed`: a subfield of $\mathbb{R}$ closed under taking real roots of its polynomials is real closed; in particular the relative algebraic closure `relAlgebraic K₀` of any subfield $K_0 \subseteq \mathbb{R}$ is real closed.

Theorem. `exists_rc_subfield`: for every cardinal κ with $\aleph_0 \leq \kappa \leq c$, there exists a real-closed subfield of $\mathbb{R}$ of cardinality exactly κ .

Remark. This proves (and strengthens) the former `subfieldStratumExists` axiom. The subfield is the relative algebraic closure of the subfield generated by a set of size κ ; the cardinality is pinned by `Algebra.IsAlgebraic.cardinalMk_le_max` for the upper bound and injectivity of the generators for the lower bound.

4.5. Real algebraic numbers as a base field.

Theorem. `algReal_card_le_aleph0`: the real algebraic numbers, represented as `algebraicClosure QQ RR` inside $\mathbb{R}$, have cardinality at most $\aleph_0$.

Together with the lower bound for infinite types, this supports the concrete base object `algRealSAF`, a countable size-aware field outside the intermediate-cardinality hierarchy.

4.6. Definability of addition and multiplication.

Theorem. `graphDefinable_add` and `graphDefinable_mul`: the graphs of addition and multiplication on $\mathbb{R}$ are first-order definable in the selected ring language with parameters.

These lemmas are technically demanding because the proof must build explicit bounded formulas and then verify realization through several layers of Mathlib's model-theory API.

4.7. Elementarity-preserving composition.

Theorem. `embLE`: for any $m \leq n$, the chain induces an elementary embedding from level m to level n .

Theorem. `embLE_eq_sysEmb`: the elementarity-preserving embedding built by recursive composition agrees as a function with the directed-system embedding used by `Language.DirectLimit`.

4.8. Łoś-style theorem for direct limits.

Theorem. `ElemChain.losDirectLimit`: if every level of an elementary chain embeds elementarily and compatibly into $\mathbb{R}$, then the direct limit is elementarily equivalent to $\mathbb{R}$. Formerly an axiom.

Remark. The underlying map is built by `Language.DirectLimit.lift`; elementarity is established by the Tarski–Vaught test: a witness of an existential in $\mathbb{R}$ over parameters from the limit is pulled back to a finite level via the compatibility of the embeddings.

4.9. Cardinality of the direct limit.

Theorem. `directLimit_card`: if every level has cardinality below continuum and the supremum of the level cardinalities is continuum, then the direct limit has cardinality continuum.

The proof has two halves. The upper bound exhibits the direct limit as a quotient of the sigma type $\sum_{n:\mathbb{N}} C_n$, bounding its cardinality by $\aleph_0 \cdot c = c$. The lower bound uses injectivity of the canonical maps from each level into the direct limit and then takes the supremum over levels.

4.10. Limit size via the cofinal family.

Theorem. `exists_cofinal_rc_family` and `cofinal_family_limit_size`: there is a monotone, $\mathbb{Q}$.ord-indexed family of real-closed subfields of $\mathbb{R}$, each of intermediate cardinality under $\neg \text{CH}$, whose union is all of $\mathbb{R}$ and hence has cardinality c .

Remark. The shift from an $\mathbb{N}$ -index to the ordinal $\mathbb{Q}$.ord is forced by König’s theorem: $\text{cof}(c) > \aleph_0$, so no countable chain of strata of size $< c$ can be cofinal in $\mathbb{R}$. This corrected formulation is what `ICAHStatement` now asserts for the limit-size clause.

Contribution. For the paper, `losDirectLimit`, `directLimit_card`, and `exists_rc_subfield` are the strongest fully proved theorems to foreground. Each is independent of the grand interpretation of ICAH: the first two are reusable model-theoretic and cardinal-arithmetic facts about `Language.DirectLimit`, and the third produces real-closed subfields of $\mathbb{R}$ of every infinite cardinality up to c .

5. AXIOM INVENTORY AND MATHLIB ROADMAP

A central strength of the Lean development is that it does not hide its assumptions. The theorem `icahTheorem` is assembled from proved components plus a small set of named dependencies. After the June 2026 axiom-reduction effort, the inventory contains exactly **two** project axioms, enforced in-source by a `#guard_msgs` in `#print axioms icahTheorem` check and audited in CI:

```
'ICAH.icahTheorem' depends on axioms:
  [propext, Classical.choice, not_CH, rcfModelComplete, Quot.sound]
```

5.1. External mathematical assumption.

Assumption. `ICAH.not_CH`: the negation of the Continuum Hypothesis, represented as $\text{continuum} \neq \aleph_1$.

This is not a Mathlib gap. It is the intended set-theoretic regime of the project: the theory is developed relative to $\neg \text{CH}$.

5.2. The single remaining Mathlib gap.

Mathlib gap. `ICAH.rcfModelComplete`: model completeness of the theory of real-closed ordered fields (Tarski–Seidenberg). Concretely: the inclusion of a real-closed subfield of $\mathbb{R}$ into $\mathbb{R}$ is an elementary embedding in the language of ordered rings.

This is a true classical theorem whose first-order formalization (quantifier elimination or model completeness for RCF inside Mathlib’s `ModelTheory` framework) is absent from Mathlib. It is the natural next contribution target, and discharging it would reduce the project to the single definitional axiom `not_CH`.

5.3. Resolved former axioms.

The five remaining entries of the previous inventory were all discharged:

Theorem. `ICAH.Real.isRealClosed` — **proved** (`ICAH/RealClosed.lean`): $\mathbb{R}$ is real closed, via `IsRealClosed.of_linearOrderedField`, `Real.sqrt` for squares, and the intermediate value theorem for odd-degree polynomials.

Theorem. `ICAH.fieldOnStratum` — **proved** (`ICAH/FieldOnStratum.lean`): every stratum admits a size-aware field, by `Equiv-transport` of the field structure of a real-closed subfield of matching cardinality.

Theorem. `ICAH.subfieldStratumExists` — **proved and strengthened** (`exists_rc_subfield`): for every $\aleph_0 \leq \kappa \leq c$ there is a **real-closed** subfield of $\mathbb{R}$ of cardinality exactly κ , obtained as the relative algebraic closure of a generated subfield.

Theorem. `ICAH.ElemChain.losDirectLimit` — **proved** (`ICAH/ElementaryChain.lean`): the direct limit of an elementary chain compatibly embedded in $\mathbb{R}$ is elementarily equivalent to $\mathbb{R}$, via `Language.DirectLimit.lift` and the Tarski–Vaught test.

Remark. Two further axioms of the earlier draft, `subfieldIsRealClosed` and `subfieldStratumElemEmb`, were **deleted as false in the stated generality**: $\mathbb{Q}$ is a subfield of $\mathbb{R}$ that is neither real closed nor elementarily embedded. They are replaced by the sound refinement `RCSubfieldStratum` (a stratum whose carrier is a real-closed subfield), for which elementarity follows from `rcfModelComplete`.

5.4. The honest limit construction.

The original $\mathbb{N}$ -indexed chain formulation of the limit-size milestone is vacuous under $\neg \text{CH}$: König's theorem gives $\text{cof}(c) > \aleph_0$, so no countable chain of intermediate-size strata can union to $\mathbb{R}$. The corrected statement (`ICAH/CofinalFamily.lean`) indexes the family by the ordinal `ℵ.ord`:

Theorem. `exists_cofinal_rc_family`: there is a monotone family of real-closed subfields of $\mathbb{R}$, indexed by `ℵ.ord`, each of intermediate cardinality, whose union is all of $\mathbb{R}$; and `cofinal_family_limit_size`: the union has cardinality c .

5.5. Recommended order of attack.

1. Extract `IsRealClosed` `RR` and the root-closure criterion `isRealClosed_of_forall_root` as Mathlib PRs.
2. Extract the elementary-direct-limit theorem (`losDirectLimit`) for `Language.DirectLimit` as a Mathlib PR.
3. Formalize quantifier elimination / model completeness for RCF in Mathlib's `ModelTheory` framework, discharging `rcfModelComplete` — the final gap.

6. RELATED WORK

6.1. Continuum hypothesis and independence.

The set-theoretic background is classical. The Continuum Hypothesis asks whether $2^{\aleph_0} = \aleph_1$, equivalently whether there is no intermediate cardinal between $\aleph_0$ and the continuum. Gödel proved relative consistency of CH with ZFC, and Cohen proved the independence direction using forcing [1, 2].

The present work does not contribute to the independence problem itself. It works in the $\neg \text{CH}$ regime and asks what kind of algebraic and model-theoretic structure can be layered over intermediate-cardinality subsets or subfields of $\mathbb{R}$.

6.2. Real closed fields.

The algebraic and model-theoretic backbone is the theory of real closed fields. Tarski's quantifier elimination for real closed fields makes them a natural setting for definability questions involving addition, multiplication, and order [3, 4].

ICAH uses this tradition, but the novelty of the formalization is to combine real-closed-field ideas with explicit cardinal bookkeeping and Lean-level axiom accounting.

6.3. Formalized mathematics in Lean.

The project belongs to the growing ecosystem of Lean 4 and Mathlib formalizations. Mathlib is a community-driven library of formalized mathematics used for research-scale formal proofs [6]. ICAH is not only a Lean artifact but also a

diagnostic of Mathlib boundaries: it identifies concrete places where model theory, real closed fields, cardinal arithmetic, and directed colimits need stronger APIs.

6.4. Direct limits and elementary chains.

Elementary-chain arguments are standard in model theory. The formal contribution here is more specific: it tests how far Mathlib’s `Language.DirectLimit` API can be pushed toward a fully formal elementary-chain theorem. Both the direct-limit cardinality result and the elementarity theorem (`losDirectLimit`, via `DirectLimit.lift` and the Tarski–Vaught test) are now proved locally; both are natural candidates for upstreaming to Mathlib.

7. NEIGHBORING FIELDS AND ANALOGUES

The ICAH architecture has natural analogues beyond real closed fields. The closest comparisons are not arbitrary fields but fields equipped with a closure principle and a first-order theory.

7.1. P-adically closed fields.

P-adically closed fields are the valued-field analogue most similar in spirit to real closed fields. On the real side, order and real closure control definability. On the p-adic side, valuation, henselianity, and p-adic closure play the corresponding role. Model theory of valued fields studies exactly this pattern: a field is enriched by extra structure, and the central questions concern definability, elementary equivalence, and preservation under extensions or chains [7, 8].

A p-adic analogue of ICAH would replace:

1. subsets or subfields of $\mathbb{R}$ by valued subfields of a p-adic ambient field;
2. order/real-closure hypotheses by valuation/henselian or p-adically-closed hypotheses;
3. definability in an ordered-ring language by definability in a valued-field language;
4. real-closed-field elementary chains by valued-field elementary chains.

7.2. Henselian valued fields.

Henselian valued fields provide a broader comparison class. They are not merely fields with a topology; they are algebraic structures with a valuation satisfying a lifting principle. This makes them suitable for direct analogues of the ICAH pattern: carrier, extra structure, closure property, definability theory, elementary embeddings, and direct limits.

7.3. Algebraically closed fields.

Algebraically closed fields are less close to the order-theoretic content of ICAH, but they share an important formal pattern: a strong closure property plus a well-behaved first-order theory. A stripped-down version of the ICAH architecture could be tested in algebraically closed fields as a simpler control case.

7.4. Paper positioning.

Remark. The safest comparison is: ICAH is a real-field, cardinal-aware, Lean-formalized instance of a broader model-theoretic pattern. The neighboring fields are real closed fields, p-adically closed fields, henselian valued fields, pseudo real closed fields, pseudo p-adically closed fields, and algebraically closed fields.

For the first paper, these analogies should remain in a future-work section. The core contribution should stay focused on the formal Lean development around the real continuum.

8. CONCLUSION

The current ICAH formalization is best presented as a rigorous formalization study rather than as a finished foundational theory. Its strongest mathematical components are already useful beyond ICAH itself: the real-closedness of $\mathbb{R}$ (`Real.isRealClosed`), the existence of real-closed subfields of $\mathbb{R}$ of every infinite cardinality up to the continuum (`exists_rc_subfield`), the Łoś-style elementarity theorem for direct limits (`losDirectLimit`), the cardinality theorem for direct limits (`directLimit_card`), and the continuum-indexed cofinal family of real-closed subfields (`exists_cofinal_rc_family`).

The most important editorial decision is to keep the paper honest about the distinction between proved Lean results and named assumptions. This is not a weakness. It is one of the project’s strongest features: the axiom inventory — now reduced to $\neg \text{CH}$ plus the model completeness of real-closed fields, and machine-checked by a `#guard_msgs` audit in the build — turns an ambitious mathematical program into one precise, independently attackable formalization problem.

8.1. Immediate next steps.

1. Formalize quantifier elimination or model completeness for real-closed fields in Mathlib’s `ModelTheory` framework, discharging `rcfModelComplete` and reducing the project to the single definitional axiom $\neg \text{CH}$.
2. Upstream the reusable components as Mathlib PRs: `Real.isRealClosed`, the root-closure criterion `isRealClosed_of_forall_root`, `losDirectLimit`, and `directLimit_card`.
3. Paste the exact `#print axioms icahTheorem` output into an appendix (currently: `propext`, `Classical.choice`, `not_CH`, `rcfModelComplete`, `Quot.sound`).
4. Decide whether the first submission target is a formalization venue, a Lean/Mathlib note, or a broader mathematical logic preprint.
5. Keep p-adic and valued-field analogies as future work, not as the main thesis.

TODO. Before public release, replace the informal claim about the remaining Mathlib gap (`rcfModelComplete`) with either a direct link to the relevant Mathlib discussion or a minimized Lean example showing the missing theorem.

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